CS6650 Assignment 2: Message Distribution and Queue Management

Student Name: Suiyang Mai
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1. Git Repository URL

Repository: `https://github.com/maisuiyang/cs6650`

Project Structure

- `/server-v2` - WebSocket server with RabbitMQ queue integration
- `/consumer` - Consumer application (queue to broadcast)
- `/deployment` - ALB configuration and deployment scripts
- `/monitoring` - Monitoring scripts and tools
- `/results` - Test results and screenshots

See README in each directory for running instructions.

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2. Architecture Document

2.1 System Architecture

```
Client (send) --> server-v2:8081 (WebSocket)
                        |
                        v
                  RabbitMQ
                  chat.exchange (topic)
                  room.1 ... room.20 (queues)
                        |
                        v
                  Consumer:8082 (consumes & broadcasts)
                        |
                        v
                  Client (receive) <-- WebSocket
```

- **server-v2**: Accepts WebSocket messages, validates JSON, publishes to RabbitMQ (topic exchange `chat.exchange`, routing key `room.{roomId}`), sends ACK to sender.
- **RabbitMQ**: Topic exchange with 20 queues (`room.1` through `room.20`). One queue per room for ordered delivery per room.
- **consumer**: Consumes from all 20 queues, maintains per-room WebSocket sessions, broadcasts each message to all clients connected to that room on port 8082.

2.2 Message Flow

1. Client connects to server-v2 (`ws://host:8081/chat/{roomId}`) to send messages.
2. Client connects to consumer (`ws://host:8082/chat/{roomId}`) to receive broadcasts.
3. On send: server-v2 validates message, publishes to exchange with routing key `room.{roomId}`, responds with status OK and messageId.
4. Message lands in queue `room.{roomId}`. Consumer consumes it, broadcasts JSON to all sessions in that room.
5. All clients in the room receive the same message (group chat).

2.3 Queue Topology

- **Exchange:** `chat.exchange` (topic, durable).
- **Queues:** `room.1` … `room.20` (classic, durable).
- **Bindings:** Each queue bound with routing key `room.1` … `room.20` respectively.

2.4 Consumer Threading Model

- Spring AMQP listener concurrency: 20 (configurable via `spring.rabbitmq.listener.simple.concurrency`).
- Single `@RabbitListener` subscribes to all 20 queues. One thread pool handles all queues; messages are processed in order per queue.
- `RoomSessionManager`: `ConcurrentHashMap<roomId, Set<WebSocketSession>>` for thread-safe add/remove/broadcast.

2.5 Load Balancing

- ALB in front of multiple server-v2 instances. Target group health check: `/health`, 30s interval, 5s timeout.
- Sticky sessions enabled so each WebSocket connection stays on the same server-v2 instance.
- Idle timeout set to 60s or higher for WebSocket.

2.6 Failure Handling

- server-v2: On RabbitMQ publish failure, client can receive error or retry; connection pooling and channel reuse reduce failures.
- consumer: On broadcast failure (e.g. closed session), message is still acked so it is not redelivered; other sessions in the room receive the message.
- Queue unavailability: Spring AMQP retries and reconnects; circuit breaker can be added for production.

3. Test Results

3.1 Single Instance (AWS)

Environment: AWS us-west-2; 1× server-v2 behind ALB, RabbitMQ and consumer on separate EC2; client from local Mac.

Client Output (32 threads)

Warmup (32,000 messages)

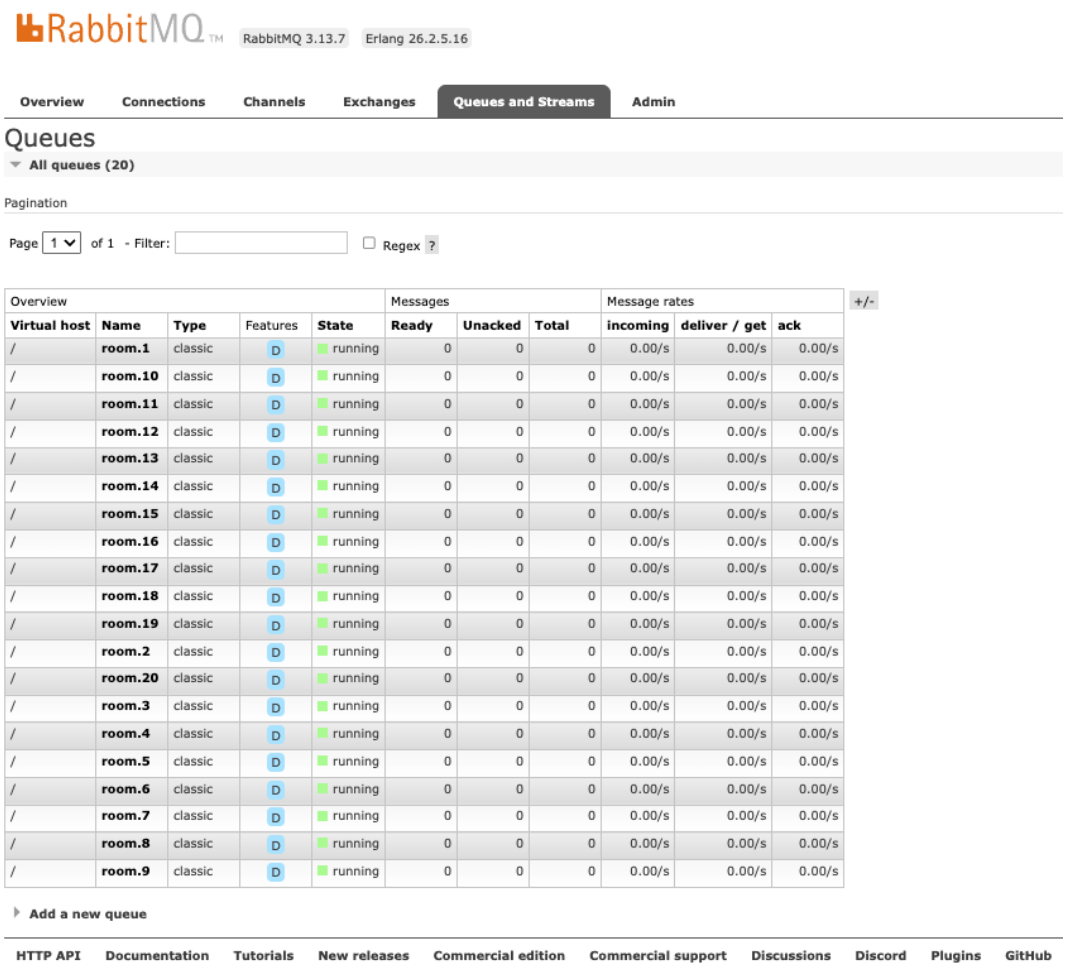
- Wall time: 6.71 s
- Throughput: 4,772 msg/s
- Success: 32,000, Failed: 0

Main (500,000 messages)

- Wall time: 19.00 s
- Throughput: 26,313 msg/s
- Success: 500,000, Failed: 0

```
===== WARMUP START =====
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See https://www.slf4j.org/codes.html#noProviders for further details.
===== WARMUP RESULT =====
Success: 32000
Failed: 0
Retries: 0
Wall time (s): 6.706
Throughput (msg/s): 4771.846107963018
Connections: 622
Reconnections: 622
CSV saved to: results/WARMUP-latency.csv
Latency count: 0
mean(ms): 0
median(ms): 0
p50(ms): 0
p95(ms): 0
p99(ms): 0
min(ms): 0
max(ms): 0
Throughput per room (msg/s):
Message type distribution:
===== WARMUP END =====
===== MAIN START =====
===== MAIN RESULT =====
Success: 500000
Failed: 0
Retries: 0
Wall time (s): 19.002
Throughput (msg/s): 26313.019682138725
Connections: 647
Reconnections: 647
CSV saved to: results/MAIN-latency.csv
Latency count: 0
mean(ms): 0
median(ms): 0
p50(ms): 0
p95(ms): 0
p99(ms): 0
min(ms): 0
max(ms): 0
Throughput per room (msg/s):
Message type distribution:
===== MAIN END =====
```

- During and after the run, queue depths (Ready / Total) for `room.1`–`room.20` remained at 0, indicating the consumer kept up with the producer with no backlog.



Overview					Messages			Message rates		
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack
/	room.1	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.10	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.11	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.12	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.13	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.14	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.15	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.16	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.17	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.18	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.19	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.2	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.20	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.3	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.4	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.5	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.6	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.7	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.8	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s
/	room.9	classic	D	running	0	0	0	0.00/s	0.00/s	0.00/s

3.2 Load Balanced (2 instances)

Environment: Same as 3.1; 2× server-v2 (server1, server2) in target group behind ALB.

Client Output (500K messages)

Warmup (32,000 messages)

- Wall time: 4.45 s
- Throughput: 7,196 msg/s
- Success: 32,000, Failed: 0
- Latency: mean 1190 ms, median 1250 ms, p95 2182 ms, p99 2349 ms

Main (500,000 messages)

- Wall time: 14.84 s
- Throughput: **33,695 msg/s**
- Success: 500,000, Failed: 0
- Latency: mean 4592 ms, median 4924 ms, p95 9029 ms, p99 9489 ms

```
===== WARMUP START =====
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See https://www.slf4j.org/codes.html#noProviders for further details.
===== WARMUP RESULT =====
Success: 32000
Failed: 0
Retries: 0
Wall time (s): 4.447
Throughput (msg/s): 7195.862379131999
Connections: 531
Reconnections: 531
CSV saved to: results/WARMUP-latency.csv
Latency count: 14248
mean(ms): 1190
median(ms): 1250
p50(ms): 1250
p95(ms): 2182
p99(ms): 2349
min(ms): 20
max(ms): 3407
Throughput per room (msg/s):
room 1: 255.67798515853383
room 2: 84.77625365414886
room 3: 178.5473352822127
room 4: 25.410389026309872
room 5: 255.22824375983808
room 6: 197.88621542612998
room 7: 248.70699347874972
room 8: 185.74319766134474
room 9: 229.36811333483246
room 10: 191.81470654373734
room 11: 84.77625365414886
room 12: 88.37418484371486
room 13: 24.960647627614122
room 14: 221.9473802563526
room 15: 185.9680683606926
room 16: 23.386552732179
room 17: 14.841466156959749
room 18: 236.56397571396445
room 19: 223.29660445243985
room 20: 246.68315718461884
Message type distribution:
JOIN: 727
LEAVE: 703
TEXT: 12818
===== WARMUP END =====
```

```
===== MAIN START =====
===== MAIN RESULT =====
Success: 500000
Failed: 0
Retries: 0
Wall time (s): 14.839
Throughput (msg/s): 33694.99292405148
Connections: 642
Reconnections: 642
CSV saved to: results/MAIN-latency.csv
Latency count: 61514
mean(ms): 4592
median(ms): 4924
p50(ms): 4924
p95(ms): 9029
p99(ms): 9489
min(ms): 21
max(ms): 10034
Throughput per room (msg/s):
room 1: 173.79877350225755
room 2: 195.76790888873913
room 3: 222.6565132421322
room 4: 215.17622481299279
room 5: 229.5976817844868
room 6: 112.87822629557247
room 7: 198.39611833681514
room 8: 183.23337152099197
room 9: 185.12029112473886
room 10: 220.63481366668913
room 11: 232.09111126086663
room 12: 199.00262820944806
room 13: 240.4474695060314
room 14: 256.2841161803356
room 15: 227.4412022373475
room 16: 193.67881932744794
room 17: 228.45205202506907
room 18: 220.56742368084102
room 19: 194.28532920008087
room 20: 215.9175146573219
Message type distribution:
JOIN: 3129
LEAVE: 3059
TEXT: 55326
===== MAIN END =====
```

Performance vs single instance: Throughput increased from ~26,313 msg/s to ~33,695 msg/s (~28% improvement); wall time for Main reduced from 19.0 s to 14.8 s. Queue depths for `room.1`–`room.20` remained low (near 0), similar to the single-instance run.

3.3 Load Balanced (4 instances)

Environment: Same; 4× server-v2 (server1–server4) in target group behind ALB.

Client Output (500K messages)

Warmup (32,000 messages)

- Wall time: 3.80 s
- Throughput: 8,414 msg/s
- Success: 32,000, Failed: 0
- Latency: mean 848 ms, median 1036 ms, p95 1742 ms, p99 1960 ms

Main (500,000 messages)

- Wall time: 10.87 s
- Throughput: **45,985 msg/s**
- Success: 500,000, Failed: 0
- Latency: mean 2200 ms, median 1915 ms, p95 5380 ms, p99 6276 ms

```
===== WARMUP START =====
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See https://www.slf4j.org/codes.html#noProviders for furt
her details.
===== WARMUP RESULT =====
Success: 32000
Failed: 0
Retries: 0
Wall time (s): 3.803
Throughput (msg/s): 8414.409676571127
Connections: 585
Reconnections: 585
CSV saved to: results/WARMUP-latency.csv
Latency count: 22266
mean(ms): 848
median(ms): 1036
p50(ms): 1036
p95(ms): 1742
p99(ms): 1960
min(ms): 17
max(ms): 2119
Throughput per room (msg/s):
room 1: 372.07467788587957
room 2: 316.5921640809887
room 3: 330.2655798054168
room 4: 269.7870102550618
room 5: 318.69576650013147
room 6: 360.2419142782014
room 7: 292.66368656323954
room 8: 128.3197475677097
room 9: 231.39626610570602
room 10: 424.13883775966343
room 11: 369.7081251643439
room 12: 417.5650801998422
room 13: 275.04601630291876
room 14: 107.02077307388903
room 15: 308.96660531159614
room 16: 162.24033657638708
room 17: 364.449119116487
room 18: 185.90586379174337
room 19: 401.7880620562714
room 20: 217.9858006836708
Message type distribution:
JOIN: 1066
LEAVE: 1108
TEXT: 20092
===== WARMUP END =====
```

```
===== MAIN START =====
===== MAIN RESULT =====
Success: 500000
Failed: 0
Retries: 0
Wall time (s): 10.873
Throughput (msg/s): 45985.46859192495
Connections: 640
Reconnections: 640
CSV saved to: results/MAIN-latency.csv
Latency count: 103606
mean(ms): 2200
median(ms): 1915
p50(ms): 1915
p95(ms): 5380
p99(ms): 6276
min(ms): 19
max(ms): 8067
Throughput per room (msg/s):
room 1: 413.3173917042215
room 2: 602.6855513657684
room 3: 435.48238756552934
room 4: 653.5454796284374
room 5: 219.25871424629818
room 6: 439.8969925503541
room 7: 517.6124344707073
room 8: 442.74809160305347
room 9: 573.8986480272234
room 10: 459.2108893589626
room 11: 382.13924399889635
room 12: 470.0634599466569
room 13: 546.2153959348846
room 14: 578.5891658235997
room 15: 569.3920721052148
room 16: 551.9175940402833
room 17: 469.5116343235538
room 18: 326.588797939851
room 19: 332.65887979398514
room 20: 544.0080934424722
Message type distribution:
JOIN: 5314
LEAVE: 5350
TEXT: 92942
===== MAIN END =====
```

Performance comparison: Throughput improved from ~26,313 msg/s (1 instance) and ~33,695 msg/s (2 instances) to ~45,985 msg/s (4 instances), while keeping error rate at 0; latency increases but remains acceptable for this workload, indicating near-linear scaling with four servers. Queue depths across all rooms stayed well below 1000 messages in all three scenarios, so the system maintained stable queues under load.

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4. Configuration Details

Component	Value
Queue	Exchange `chat.exchange` (topic, durable). Queues `room.1`–`room.20`. Routing key `room.{roomId}`.
Consumer	Port 8082. Concurrency 20. Prefetch 50.
server-v2	Port 8081. RabbitMQ host/port via `SPRING_RABBITMQ_HOST`.
ALB	Health `/health`, 30s interval, 5s timeout. Sticky sessions on. Idle timeout 60s+. Listener HTTP:80.
Instances	t2.micro, us-west-2. Target group hw2-websocket-servers (port 8081).